

MINI SHOP MANAGEMENT SYSTEM

Python Assignment 2 | Program Demonstration Report

This report presents a complete walkthrough of the Mini Shop Management System, a menu-driven Python program built to help a small shop owner manage products, track stock, process sales, and view business summaries. Each section below shows a screenshot of a specific feature in action, captured directly from the program's console output.

#	Feature Demonstrated
1	Main Menu
2	Adding Products
3	Viewing Products
4	Searching for a Product
5	Selling a Product (incl. stock checks)
6	Low Stock Report
7	Sales Summary
8	Bonus: Sort by Price & Search by Name
9	Input Validation & Error Handling
10	Program Exit (with Recursion Check)

1. Main Menu

```
=====
      MINI SHOP SYSTEM
=====

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice: 1
Enter Product ID: 123
Enter Product Name: Glass
Enter Price: 50
Enter Quantity: 6
Choose a category:
    1. Stationery
    2. Electronics
    3. Grocery
Enter category number: 3
Product added successfully!
```

The program displays a numbered menu with all available options (Add, View, Search, Sell, Low Stock, Sales Summary, bonus features, and Exit). The menu keeps showing after every action until the user chooses to exit.

2. Adding Products

Product 1: Adding "Glass" (ID: 123)

```
=====
      MINI SHOP SYSTEM
=====

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice: 1
Enter Product ID: 123
Enter Product Name: Glass
Enter Price: 50
Enter Quantity: 6
Choose a category:
  1. Stationery
  2. Electronics
  3. Grocery
Enter category number: 3
Product added successfully!
```

Product 2: Adding "Pens" (ID: 124)

```
=====
      MINI SHOP SYSTEM
=====

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice: 1
Enter Product ID: 124
Enter Product Name: Pens
Enter Price: 10
Enter Quantity: 100
Choose a category:
  1. Stationery
  2. Electronics
  3. Grocery
Enter category number: 1
Product added successfully!
```

The shopkeeper enters a Product ID, Name, Price, Quantity, and chooses a category. The product is then stored and a confirmation message is shown.

3. Viewing Products

Enter your choice: 2

===== PRODUCTS =====

ID: 123
Name: Glass
Price: Rs: 50.0
Stock: 6
Category: Grocery

ID: 124
Name: Pens
Price: Rs: 10.0
Stock: 100
Category: Stationery

=====

MINI SHOP SYSTEM

=====

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice:

All products currently in the shop are listed with their ID, name, price, stock, and category, displayed using a for loop that iterates through the product dictionary.

4. Searching for a Product

Search Result — Product Found (ID: 123)

```
=====
      MINI SHOP SYSTEM
=====

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice: 3
Enter Product ID: 123

Product Found!
Name: Glass
Price: Rs. 50.0
Stock: 6
```

Search Result — Product Not Found (ID: 134)

```
=====
      MINI SHOP SYSTEM
=====

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice: 3
Enter Product ID: 134
Product not found!
```

The `search_product()` function checks the entered Product ID against the stored records and returns the product's details if found, or a 'Product not found!' message otherwise.

5. Input Validation & Error Handling

Duplicate Product ID Warning

```
=====
      MINI SHOP SYSTEM
=====

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice: 1
Enter Product ID: 123
Warning: A product with this ID already exists!
```

Invalid Menu Choice

```
=====
      MINI SHOP SYSTEM
=====

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice: 123
Invalid choice, please try again.
```

Invalid Price Entry (non-numeric input)

```
=====
      MINI SHOP SYSTEM
=====
```

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice: 1

Enter Product ID: 134

Enter Product Name: TV

Enter Price: Ninety five thousand

Error: Invalid price or quantity. Product not added.

The program gracefully handles invalid or unexpected input — duplicate product IDs, out-of-range menu choices, and non-numeric price/quantity entries — without crashing.

6. Selling a Product

Attempt to Search an Unavailable Product Before Sale (ID: 135)

```
=====
      MINI SHOP SYSTEM
=====

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice: 3
Enter Product ID: 135
Product not found!
```

Successful Sale — Bill Generated for Glass (Qty: 3)

```
=====
      MINI SHOP SYSTEM
=====

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice: 4
Enter Product ID: 123
Enter Quantity: 3

===== BILL =====
Product: Glass
Quantity: 3
Price: Rs: 50.0
Total: Rs: 150.0
Sale completed!
```

6. Selling a Product (continued)

Insufficient Stock Check (requesting more than available)

```
=====
      MINI SHOP SYSTEM
=====

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice: 4
Enter Product ID: 123
Enter Quantity: 4
Error: Not enough stock! Only 3 left.
```

Invalid Quantity Check (negative quantity entered)

```
=====
      MINI SHOP SYSTEM
=====

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice: 4
Enter Product ID: 124
Enter Quantity: -2
Error: Quantity must be greater than zero.
```

Before completing a sale, the program verifies that the product exists, the quantity is valid, and enough stock is available. On success, it reduces the stock and prints a bill with the total price.

7. Low Stock Report

```
=====
      MINI SHOP SYSTEM
=====

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice: 5

LOW STOCK PRODUCTS
Glass - 3 remaining
```

Using a list comprehension, the program filters out and displays all products with stock below 5 units, warning the shopkeeper to restock soon.

8. Sales Summary

```
=====
      MINI SHOP SYSTEM
=====
```

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice: 6

```
===== SALES SUMMARY =====
Total Products Sold: 3
Total Sales Amount: Rs. 150.0
```

A running total of all completed sales is displayed — total items sold and total revenue earned — calculated from the list of recorded sales.

9. Bonus Features

Sort Products by Price (Lambda Function, Low to High)

```
=====
      MINI SHOP SYSTEM
=====

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice: 7

PRODUCTS SORTED BY PRICE (Low to High)
Pens - Rs. 10.0
Glass - Rs. 50.0
```

Search Products by Name (Partial Match)

```
=====
      MINI SHOP SYSTEM
=====

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price (bonus)
8. Search by Name (bonus)
9. Exit

Enter your choice: 8
Enter part of the product name: Pens

SEARCH RESULTS
124: Pens - Rs. 10.0
```

As an extra feature beyond the core requirements, the shopkeeper can sort all products by price using a lambda function, and search for products by (partial) name instead of ID.

10. Program Exit (Recursion Check)

```
=====
      MINI SHOP SYSTEM
=====
```

1. Add Product
2. View Products
3. Search Product
4. Sell Product
5. Check Low Stock
6. Sales Summary
7. Sort Products by Price
8. Search by Name
9. Exit

Enter your choice: 9

Total sale entries processed: 1

Thank you for using Mini Shop System. Goodbye!

When the shopkeeper chooses to exit, a recursive function calculates the total number of sale entries processed during the session before the program prints a goodbye message and closes.